

Controlled Data Interventions for Rural Satellite Image Segmentation

A reproducible LoveDA study of synthetic augmentation, rotation, and rare-class-aware sampling

Abstract

This study built and evaluated a complete semantic-segmentation workflow for Rural LoveDA satellite imagery. A compact U-Net was trained on 1,366 image-mask pairs and evaluated on 992 validation pairs. The main research question was whether controlled data interventions improve segmentation performance under a fixed training budget. Three interventions were tested: 10% mild brightness copies, 10% 90-degree rotation copies, and Barren-targeted class-aware oversampling. Brightness reduced clean mIoU by 0.94 percentage points on average across three seeds and did not produce a reliable robustness gain. Rotation reduced mIoU substantially in its seed-42 pilot. Class-aware oversampling raised Barren IoU from 0.00% to 3.12% on average across three seeds while leaving mIoU effectively unchanged (24.64% to 24.60%); however, pixel accuracy fell by 3.97 points. The evidence supports Barren oversampling as a targeted rare-class recall tradeoff, not as an overall performance improvement.

Research question

Can controlled synthetic image transformations or class-aware sampling improve a fixed satellite segmentation baseline?

Main conclusion

Generic brightness and rotation augmentation did not help. Barren-targeted sampling recovered an otherwise absent rare class with near-constant mIoU but lower pixel accuracy.

Plain-language summary

The model learns to color every pixel in a satellite image with a land-cover label such as road, water, forest, or agriculture. We first measured a normal training setup, then changed one thing at a time. This lets us tell whether a change actually helps instead of guessing.

1. Data and Methods

Dataset and label handling

LoveDA is a high-resolution land-cover semantic-segmentation dataset designed for rural and urban scenes [1]. This project used the Rural split only. Original PNG images and masks were 1024 x 1024 pixels; training resized both to 512 x 512. Raw LoveDA label 0 (No Data) was mapped to ignore label 255. Raw labels 1 through 7 became seven contiguous model classes: Background, Building, Road, Water, Barren, Forest, and Agriculture.

Item	Fixed setting
Training images	1,366 Rural image-mask pairs
Validation images	992 Rural image-mask pairs
Model	Compact U-Net with base channels 16 [2]
Loss	CrossEntropyLoss with ignore_index=255
Optimizer	Adam, learning rate 1e-4
Training budget	15 epochs, batch size 1, 1,366 updates per epoch
Metrics	mIoU, per-class IoU, pixel accuracy

Experimental design

A real-only baseline was trained with seeds 42, 7, and 123. Each intervention used the same architecture, split, learning rate, epochs, image size, and update budget. This paired-seed design makes comparisons fair. Best validation checkpoints were selected by mIoU.

Brightness and rotation create transformed image copies while keeping masks aligned. Barren-targeted oversampling is different: it creates no new pixels. It samples images containing Barren more often, using a weight of 4.0. Barren represented only 3.17% of training pixels.

2. Baseline and Synthetic Augmentation Results

Real-only baseline

The three-seed real-only reference averaged 24.64% validation mIoU. It achieved 0.00% Barren IoU in every seed, establishing Barren as a persistent failure class for this compact baseline.

Experiment	Evidence	Decision
10% brightness copies	Three seeds. Clean mIoU: 24.64% baseline versus 23.71% brightness. Lighting-shift changes varied by seed.	Rejected. No reliable benefit.
10% 90-degree rotations	Seed-42 pilot. mIoU fell from 24.42% to 21.54%.	Rejected early.
Barren-targeted oversampling	Three seeds. Barren IoU became nonzero in every run while mIoU stayed nearly constant.	Retained as targeted tradeoff.

Brightness robustness result

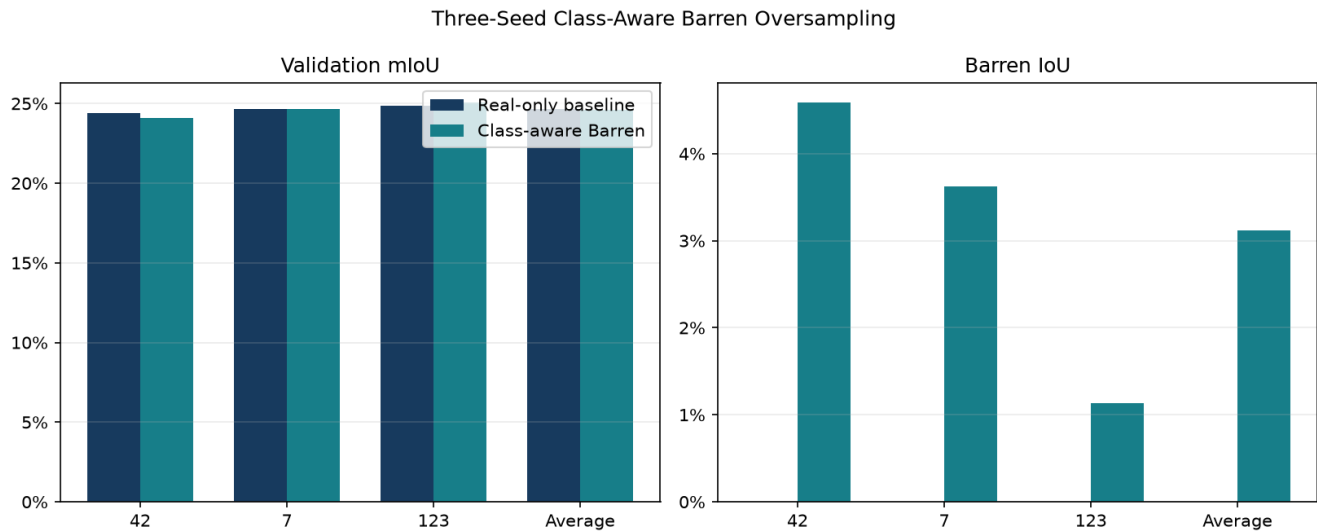
The brightness intervention was evaluated on the same validation masks under clean, darker, much darker, brighter, and much brighter images. The three-seed average showed a clean-image loss of 0.94 points. The average gains under shifted brightness were small (+0.10 to +0.59 points) and individual seeds disagreed in direction. Therefore, the study does not claim brightness augmentation improves robustness.

This negative result is valuable: it shows that an intuitive synthetic transformation should not be accepted from a single run.

3. Class-Aware Barren Oversampling

Quantitative result

Barren-targeted sampling gave Barren-containing images a fourfold sampling weight, increasing their expected presence in an epoch without changing the number of optimization updates. The chart shows all three paired seeds and their average.



Metric	Real-only average	Class-aware average	Change
Validation mIoU	24.64%	24.60%	-0.04%
Barren IoU	0.00%	3.12%	+3.12%
Pixel accuracy	53.83%	49.86%	-3.97%

Interpretation

The result is consistent across seeds in one important sense: the baseline never predicted Barren successfully, while every class-aware run produced nonzero Barren IoU. Average mIoU changed by only -0.04 points, which is negligible at this scale. Pixel accuracy fell by 3.97 points, showing that the model made more pixel-level mistakes overall while becoming more willing to identify Barren.

4. Qualitative Evaluation

What the predictions look like

The visualization below shows five validation images with ground truth, the class-aware model prediction, and an error map. It provides context that summary metrics alone cannot: the model still makes widespread confusions, particularly between visually similar land-cover regions. The class-aware method improves rare-class prediction but does not solve the broader segmentation difficulty.

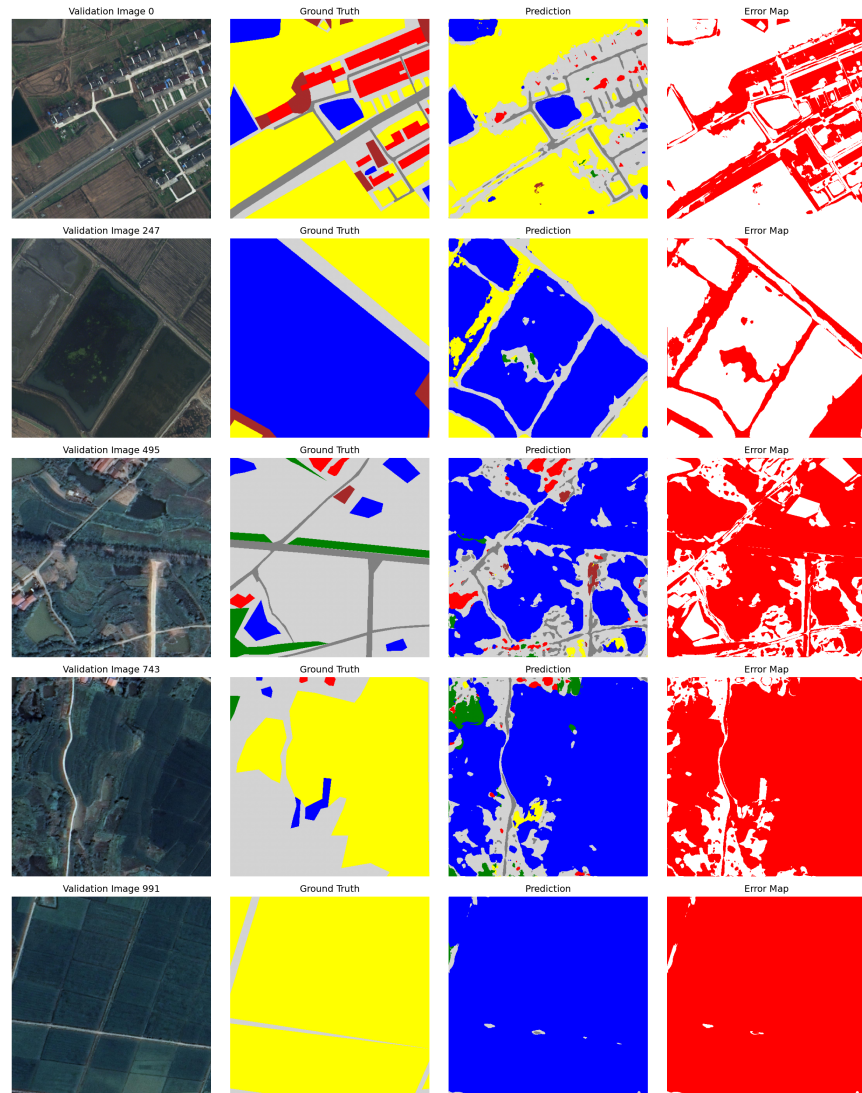


Figure 1. Validation examples for the class-aware Barren seed-42 checkpoint. Columns show the original image, ground truth, prediction, and error map.

5. Discussion and Conclusion

Answer to the research question

For this compact U-Net, data interventions did not provide a universal improvement. Mild brightness copies and rotations were not beneficial under the tested setup. Class-aware oversampling did address a specific weakness: it recovered the rare Barren class without materially changing average mIoU. The cost was lower pixel accuracy, so the preferred method depends on the project goal.

If overall pixel correctness is the priority, the real-only baseline remains the better general reference. If detecting Barren at all is important, class-aware oversampling is supported by the three repeated seeds.

Limitations

Results are limited to LoveDA Rural, one compact U-Net size, 512 x 512 input resolution, 15 epochs, and three random seeds. The study does not establish that these findings generalize to other datasets, larger architectures, or a real operational satellite system. Rotation was evaluated only in a single pilot because its decline was large. Oversampling changes the training distribution and may worsen common-class accuracy, as reflected by pixel accuracy.

Future work

A separate future study could test class-weighted loss or true copy-paste synthesis of Barren regions. Those should be treated as new questions and evaluated using the same fixed baseline and paired-seed protocol.

References

- [1] Wang, J., Zheng, Z., Ma, A., Lu, X., and Zhong, Y. LoveDA: A Remote Sensing Land-Cover Dataset for Domain Adaptive Semantic Segmentation. NeurIPS Datasets and Benchmarks, 2021. <https://arxiv.org/abs/2110.08733>
- [2] Ronneberger, O., Fischer, P., and Brox, T. U-Net: Convolutional Networks for Biomedical Image Segmentation. 2015. <https://arxiv.org/abs/1505.04597>